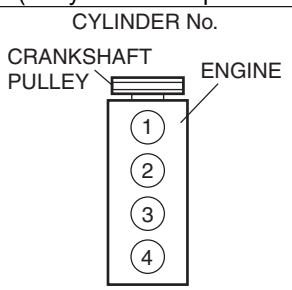
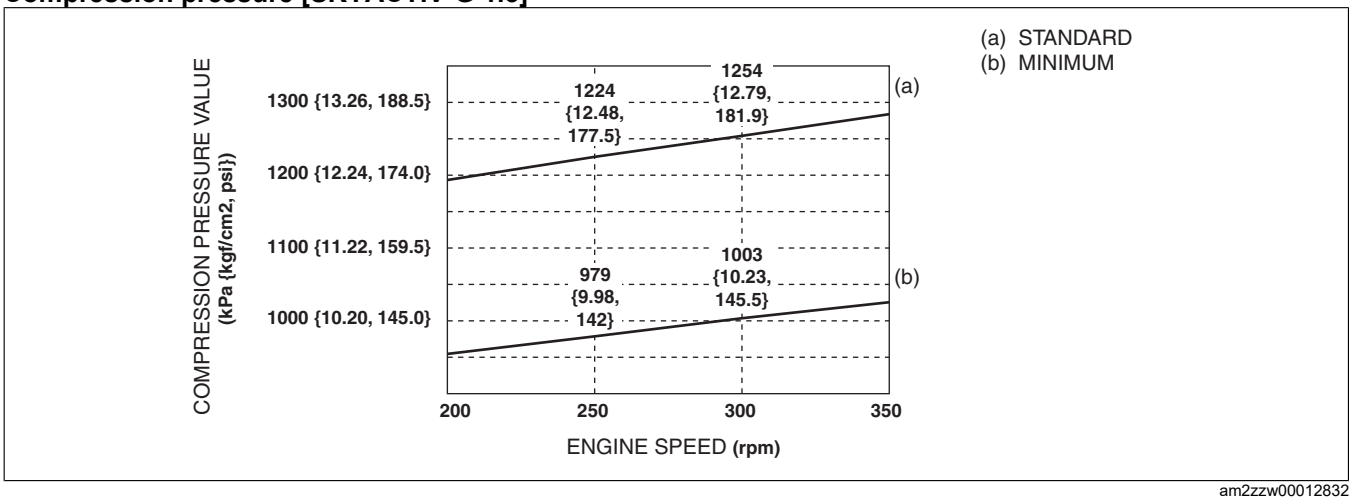
ENGINE TECHNICAL DATA [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0150g3800100

Item	Specification
Idle speed (MTX: Neutral position, ATX: P, N position)	MTX: 500—600 rpm ATX: 700—800 rpm
Ignition timing [SKYACTIV-G 1.3 (with 4-1 exhaust system)]	MTX: Approx. ATDC 3° ATX: Approx. BTDC 3°
Ignition timing [SKYACTIV-G 1.5 (with 4-1 exhaust system)]	Approx. BTDC 10°
Ignition timing [SKYACTIV-G 1.5 (with 4-2-1 exhaust system)]	Approx. BTDC 8.5°
CO concentration	Within the regulation
HC concentration	Within the regulation
Idle-up speed (MTX: Neutral position, ATX: P, N position)	MTX (A/C on (with A/C)): 650-800 rpm ATX (A/C on (with A/C)): 700-800 rpm MTX (Electrical loads on): 600-750 rpm ATX (Electrical loads on): 700-800 rpm
OCV coil resistance	6.9—7.5 ohms [20°C {68°F}]
Cylinder head bolt length L	145.2—145.8 mm {5.717—5.740 in}
Cylinder head bolt length L maximum	146.5 mm {5.767 in}
Front oil seal press on amount C	0—1.0 mm {0—0.039 in} from engine front cover
Rear oil seal press on amount C	0—0.5 mm {0—0.019 in} from cylinder block
Magnetic clutch clearance (reference)	0.3—0.6 mm {0.012—0.023 in}
Engine oil capacity (approx. quantity)	Oil replacement: 3.9 L {4.1 US qt, 3.4 Imp qt} Oil and oil filter replacement: 4.2 L {4.4 US qt, 3.7 Imp qt} Total (dry engine): 4.7 L {5.0 US qt, 4.1 Imp qt}
Oil pressure (reference value) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)] [coolant temperature: 80—90 °C {176—194 °F}]	230—400 kPa {2.35—4.07 kgf/cm^2, 33.4—58.0 psi} [3,000 rpm]
Oil pressure (reference value) [SKYACTIV-G 1.5 (with 4-2-1 exhaust system)] [coolant temperature: 80—90 ° C {176—194 °F}]	Lo: 110—175 kPa {1.13—1.78 kgf/cm^2, 16.0—25.3 psi} [1,500 rpm] Hi: 300—430 kPa {3.06—4.38 kgf/cm^2, 43.6—62.3 psi} [4,500 rpm]
Cooling system cap valve opening pressure	93.2—122.6 kPa {0.951—1.250 kgf/cm^2, 13.6—17.7 psi}
Thermostat initial-opening temperature	86.5—89.5 °C {188—193 °F}
Thermostat full-open temperature	100 °C {212 °F}
Thermostat full-open lift	More than 8.5 mm {0.33 in}
Cooling fan motor standard current [Cooling fan component type A]	7.4—11.4 A [12 V]
Cooling fan motor standard current [Cooling fan component type B]	7.0—11.0 A [12 V]
Throttle body resistance	0.3—100 ohms [20°C {68 °F}]
Fuel pressure	425—505 kPa {4.34—5.14 kgf/cm^2, 61.7—73.2 psi}
Fuel hold pressure	250 kPa {2.55 kgf/cm^2, 36.3 psi} or more
Fuel injector resistance	1.6—2.52 ohms [Surrounding temperature: 0—80 °C {32—176 °F}]
Battery electrolyte specific gravity (Without i-stop)	1.22—1.29 [20 °C {68 °F}]
Battery load test current (Without i-stop)	55D23L (60 A·h/20HR): 180 A GR35 (60 A·h/20HR): 180 A N-55 (50 A·h/20HR): 130 A
Battery parasitic draw (When the ignition is off, all doors and the bonnet are closed.)	**Vehicles without i-ELOOP** • 40—65 mA **Vehicles with i-ELOOP** • 25—45 mA
Generator standard voltage [IG-ON] (Without i-ELOOP)	Terminal 1A: B+ Terminal 2A: Approx. 1 V or less Terminal 2B: Approx. 0 V
Generator standard voltage [Idle, 20 °C {68 °F}] (Without i-ELOOP)	Terminal 1A: 13—15 V Terminal 2A: Approx. 3—8 V Terminal 2B: Turn the electrical loads (such as headlights, blower motor, rear window defogger, brake lights) on and verify that the voltage reading increases.
Generator generated current minimum value (Without i-stop)	70% of the nominal output current (nominal output current: 100 A) [Ambient temp. 20 °C {68 °F}, voltage 13.0—15.0 V, both engine and generator are hot]

Item	Specification
Generator rotor resistance (between slip rings) [20 °C {68 °F}] (Without i-ELOOP)	1.9—2.2 ohms
Generator brush length (Without i-ELOOP)	Standard: 22.5 mm {0.886 in} Minimum: 5.0 mm {0.20 in}
Generator brush spring force (Without i-ELOOP)	Standard: 4.1—5.3 N {0.42—0.54 kgf, 1.0—1.1 lbf} Minimum: 1.7 N {0.17 kgf, 0.38 lbf}
Firing order	1-3-4-2 (all cylinders independent firing) 
Spark plug type	PE5R-18-110, PE5S-18-110, P51R-18-110
Spark plug gap	Standard: 1.05—1.40 mm {0.0414—0.0551 in} New spark plug (reference): 1.05—1.15 mm {0.0414—0.0452 in}
Spark plug resistance [25°C {77 °F}]	3.0—7.5 kilohms
Starter no-load test voltage	11 V
Starter no-load test current (With i-stop)	90 A or less
Starter no-load test current (Without i-stop)	95 A or less
Starter pinion gap (Without i-stop)	0.5—2.0 mm {0.02—0.07 in}
Starter armature runout (Without i-stop)	0.1 mm {0.004 in} max.
Starter commutator diameter (Without i-stop)	Standard: 29.4 mm {1.16 in} Minimum: 28.8 mm {1.13 in}
Segment groove depth of starter commutator (Without i-stop)	Standard: 0.5 mm {0.02 in} Minimum: 0.2 mm {0.008 in}
Starter brush length (Without i-stop)	Standard: 12.3 mm {0.484 in} Minimum: 5.5 mm {0.22 in}
Starter brush spring force (Without i-stop)	Standard: 15.0—20.4 N {1.53—2.08 kgf, 3.38—4.58 lbf} Minimum: 2.75 N {0.280 kgf, 0.618 lbf}

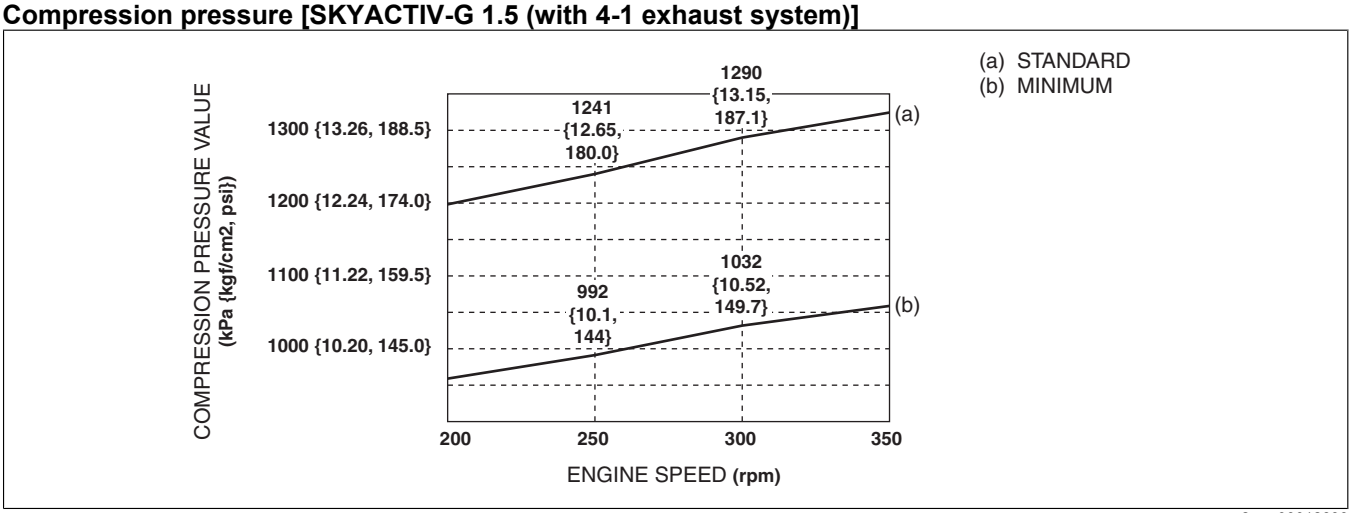
Compression pressure [SKYACTIV-G 1.3]



am2zzw00012832

Standard: Refer to the graph.
Minimum: Refer to the graph.
Maximum difference between cylinders: 208 kPa {2.12 kgf/cm^2, 30.2 psi} [250 rpm]

Compression pressure [SKYACTIV-G 1.5 (with 4-1 exhaust system)]

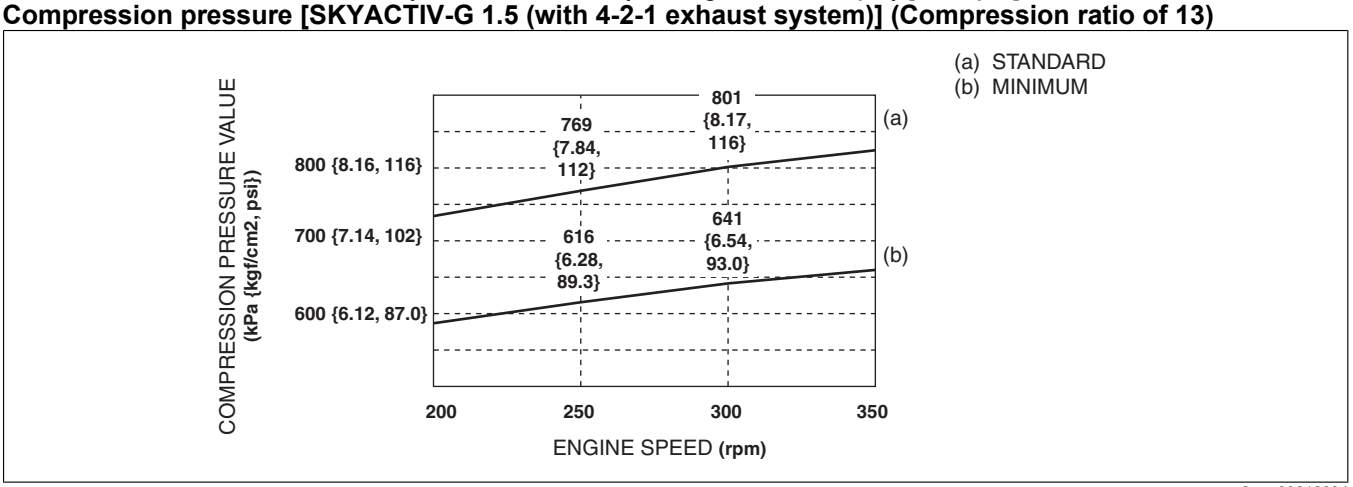


am2zzw00012833

Standard: Refer to the graph.
Minimum: Refer to the graph.

Maximum difference between cylinders: 211 kPa {2.15 kgf/cm^2, 30.6 psi} [250 rpm]

Compression pressure [SKYACTIV-G 1.5 (with 4-2-1 exhaust system)] (Compression ratio of 13)

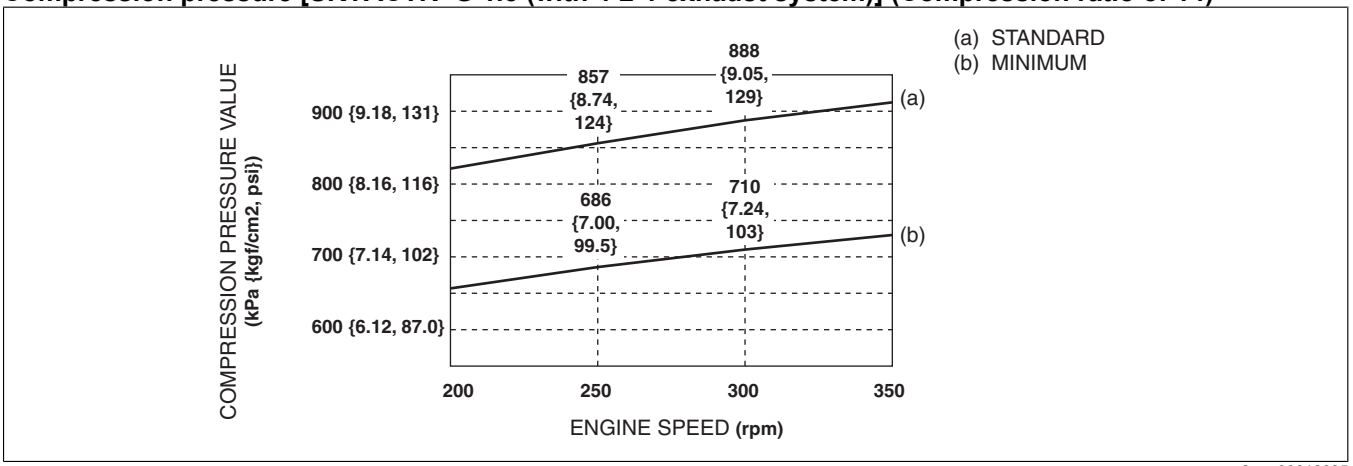


am2zzw00012834

Standard: Refer to the graph.
Minimum: Refer to the graph.

Maximum difference between cylinders: 105 kPa {1.07 kgf/cm^2, 15.2 psi} [250 rpm]

Compression pressure [SKYACTIV-G 1.5 (with 4-2-1 exhaust system)] (Compression ratio of 14)



am2zzw00012835

Standard: Refer to the graph.
Minimum: Refer to the graph.

Maximum difference between cylinders: 146 kPa {1.49 kgf/cm^2, 21.2 psi} [250 rpm]

Engine oil [Australian specs.]

Item	Specifications
Grade	API SG/SH/SJ/SL/SM/SN or ILSAC GF-2/GF-3/GF-4/GF-5

Item	Specifications
Viscosity (SAE)	10W-30, 5W-20, 5W-30, 0W-20, 0W-30

Engine oil [European (L.H.D. U.K.) specs.]

Recommended engine oil		Alternative engine oil
Mazda Original Oil Supra 0W-20	Mazda Original Oil Ultra 5W-30	API SL/SM/SN or ACEA A3/B3 or A5/B5 0W-20, 5W-30

Engine oil [Except Australian, European (L.H.D. U.K.), specs.] (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))

Item	Specifications
Grade	API SG/SH/SJ/SL/SM/SN*1 or ILSAC GF-2/GF-3/GF-4/GF-5
Viscosity (SAE)	10W-30, 5W-20, 5W-30, 0W-20, 0W-30

*1 : For African nations, use SL or higher engine oil with the SKYACTIV-G 1.3 and SKYACTIV-G 1.5 (with 4-1 exhaust system) engines.

Engine oil [Except Australian, European (L.H.D. U.K.), specs.] (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))

Item	Specifications
Grade	API SG/SH/SJ/SL/SM/SN or ILSAC GF-2/GF-3/GF-4/GF-5
Viscosity (SAE)	10W-30, 10W-40, 10W-50, 5W-20, 5W-30, 5W-40, 0W-20, 0W-30

Engine coolant capacity (approx. quantity)

Item			Capacity
European (L.H.D.), General (L.H.D.), Russia specs.		MTX	5.4 L {5.7 US qt, 4.8 Imp qt}
		ATX	5.6 L {5.9 US qt, 4.9 Imp qt}
Australian, European (R.H.D.) specs.		MTX	5.0 L {5.3 US qt, 4.4 Imp qt}
		ATX	5.2 L {5.5 US qt, 4.6 Imp qt}
General (R.H.D.) specs.(SKYACTIV-G 1.3)			
General (R.H.D.) specs.(SKYACTIV-G 1.5)	MTX	Without heater	4.5 L {4.8 US qt, 4.0 Imp qt}
		With heater	5.0 L {5.3 US qt, 4.4 Imp qt}
	ATX	Without heater	4.7 L {5.0 US qt, 4.1 Imp qt}
		With heater	5.2 L {5.5 US qt, 4.6 Imp qt}
Thailand specs.	Without heater		4.9 L {5.2 US qt, 4.3 Imp qt}
	With heater		5.4 L {5.7 US qt, 4.8 Imp qt}

Generator generated current (With i-stop, without i-ELOOP) (reference value) [Ambient temperature: 20 °C {68 °F}, Engine hot]

Engine speed (rpm)	Terminal 1A voltage (V)	Generator output current (A)
1,000	13	83
1,000	15	83
2,000	13	98
2,000	15	105

* Field coil current control signal 100%